

UST-KUT, Russian Federation

WMO: 303200

Lat: 56.857N

Lon: 105.730E

Elev: 2192

StdP: 13.57

Time Zone: 8.00 (E08)

Period: 90-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.0	-33.5	-43.1	0.5	-39.3	-36.6	0.7	-32.9	20.9	9.0	18.7	4.6	0.1	270	0.681

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	82.7	63.3	79.4	62.7	76.3	61.2	66.8	77.5	65.1	75.6	63.4	73.0	6.1	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.2	94.3	71.7	61.4	88.2	69.7	59.6	82.7	68.0	32.7	77.2	31.4	76.1	30.0	73.4	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.1	16.5	14.5	-36.3	88.1	5.9	4.4	-40.5	91.2	-44.0	93.8	-47.3	96.3	-51.5	99.5		
			-36.4	70.6	5.8	2.8	-40.6	72.6	-44.0	74.3	-47.3	75.9	-51.5	77.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	-8.7	3.8	14.6	28.8	45.1	56.6	64.9	59.3	42.8	29.2	6.8	-2.1
	DBStd	26.73	16.16	10.10	12.57	7.31	9.92	7.86	6.86	6.68	7.01	9.95	15.37	13.42
	HDD50	8896	1821	1295	1097	635	223	23	1	8	234	647	1296	1616
	HDD65	13427	2286	1715	1562	1085	620	271	89	195	666	1111	1746	2081
	CDD50	1074	0	0	0	0	73	222	462	298	19	0	0	0
	CDD65	128	0	0	0	0	4	20	84	20	0	0	0	0
	CDH74	1165	0	0	0	0	79	218	689	179	0	0	0	0
	CDH80	238	0	0	0	0	16	44	154	24	0	0	0	0
Wind	WSAvg	6.5	5.8	6.2	6.7	8.3	7.4	6.1	5.0	5.1	6.0	7.2	7.3	7.1
Precipitation	PrecAvg	19.2	1.0	0.8	0.7	0.9	1.4	2.3	2.9	3.1	1.9	1.6	1.5	1.2
	PrecMax	23.1	1.7	1.4	1.7	1.9	2.5	4.4	5.7	5.6	3.3	2.8	2.2	2.5
	PrecMin	14.2	0.3	0.4	0.1	0.4	0.4	0.7	1.0	1.8	0.2	0.5	0.9	0.6
	PrecStd	2.2	0.4	0.3	0.4	0.4	0.5	0.8	1.2	1.1	0.7	0.6	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.0	31.2	46.3	54.2	81.5	85.4	88.1	83.0	67.0	55.7	37.5	27.8
		MCWB	23.4	26.2	35.3	40.1	58.0	62.6	66.4	63.7	55.0	43.9	33.6	25.9
	2%	DB	21.0	25.2	39.8	48.4	75.1	79.8	84.4	78.6	60.7	50.3	35.3	24.3
		MCWB	20.5	22.8	32.3	37.6	54.8	61.1	64.7	63.6	51.0	40.9	32.6	22.5
	5%	DB	16.4	20.8	35.5	43.3	68.8	75.8	81.0	75.0	56.7	45.8	30.7	19.2
		MCWB	15.3	18.8	29.2	34.1	52.4	59.5	63.6	61.2	48.8	39.3	28.8	18.1
	10%	DB	12.1	16.9	31.5	39.7	62.8	71.7	77.9	71.2	53.5	41.8	26.1	15.2
		MCWB	11.5	15.4	26.8	32.5	49.6	58.6	63.2	60.1	47.1	36.6	24.4	14.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.9	26.9	35.9	41.1	60.3	67.4	69.3	68.1	55.6	44.9	34.5	25.8
		MCDB	24.8	29.9	43.5	50.6	77.3	77.8	79.9	76.8	63.6	51.0	36.5	27.5
	2%	WB	20.4	23.1	32.9	38.8	56.4	64.1	67.5	65.3	52.6	42.3	32.2	22.4
		MCDB	21.0	24.8	39.1	46.7	70.5	75.5	78.3	73.9	59.2	48.9	34.8	24.0
	5%	WB	15.3	19.0	29.9	35.9	53.4	61.5	66.1	63.2	50.1	39.9	28.9	18.3
		MCDB	16.3	20.7	34.6	41.4	67.2	71.6	77.5	71.9	54.8	45.6	30.6	19.3
	10%	WB	11.3	15.5	27.2	33.2	50.2	59.2	64.7	61.3	48.3	36.8	24.7	14.3
		MCDB	12.1	16.9	31.7	38.4	61.6	70.2	75.1	69.7	52.2	41.0	26.0	15.2
Mean Daily Temperature Range	5% DB	MDBR	11.1	14.9	18.7	14.8	19.3	18.8	18.2	17.1	13.1	12.6	11.9	12.1
		MCDBR	11.7	18.1	21.8	20.7	30.4	26.1	24.7	24.2	18.3	18.4	14.3	14.0
		MCWBR	10.8	15.6	15.9	13.4	16.2	13.2	10.3	11.5	9.9	12.2	12.6	13.8
	5% WB	MCDBR	11.1	16.8	21.2	17.4	27.6	21.2	19.4	21.0	16.0	16.7	14.3	14.5
		MCWBR	10.4	14.6	16.0	12.0	15.0	12.6	9.6	11.5	9.6	12.0	12.8	14.0
Clear-Sky Solar Irradiance	taub	0.261	0.291	0.313	0.342	0.384	0.413	0.444	0.387	0.329	0.311	0.273	0.249	
	taud	2.020	2.035	2.073	2.167	2.253	2.251	2.172	2.325	2.464	2.337	2.086	1.981	
	Ebn at Noon	187	234	264	275	270	263	250	259	262	231	181	154	
	Edh at Noon	19	30	37	40	39	40	43	34	26	22	18	15	
All-Sky Solar Radiation	RadAvg	237	542	985	1408	1642	1811	1673	1363	867	446	235	151	
	RadStd	23	39	53	98	121	141	162	92	83	51	27	21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page